Yuqian Zhang

Homepage: https://yuqianruc.github.io Email: yuqianzhang@ruc.edu.cn

Positions

Associate Professor (tenure-track) August 2026 – present
Institute of Statistics and Big Data, Renmin University of China

Assistant Professor (tenure-track) September 2022 – July 2026
Institute of Statistics and Big Data, Renmin University of China

Education

University of California, San Diego, U.S. September 2018 – March 2022
Ph.D. in Mathematics with a Specialization in Statistics
Advisor: Jelena Bradic

University of California, San Diego, U.S. September 2016 – June 2018
M.S. in Statistics

Wuhan University, China September 2012 – June 2016
B.S. in Mathematics and Applied Mathematics

Research Interests

- Causal Inference
- Semi-supervised Learning
- Dynamic Treatment Regimes
- Machine Learning and Reinforcement Learning
- High-dimensional Statistics

Publications (* corresponding, # alphabetical order, † equal contribution, __ students)

Yuqian Zhang, Weijie Ji, and Jelena Bradic (2026). Dynamic treatment effects: high-dimensional inference under model misspecification. *Journal of the Royal Statistical Society: Series B*, accepted. *(This paper received the ICSA 2025 China Conference Junior Researcher Award.)*

Yuqian Zhang†, Abhishek Chakrabortty†, and Jelena Bradic† (2026). The decaying missing-at-random framework: model doubly robust causal inference with partially labeled data. *IEEE Transactions on Information Theory*, accepted. https://doi.org/10.1109/TIT.2026.3713863.

Kai Chen and **Yuqian Zhang*** (2025). Semi-supervised linear regression: enhancing efficiency and robustness in high dimensions. *Biometrics*, 81(3), ujaf113. *(Kai received the Second Prize for Outstanding Papers at the 8th National Statistics Doctoral Academic Forum.)*

Jelena Bradic#, Weijie Ji#, and **Yuqian Zhang**# (2024). High-dimensional inference for dynamic treatment effects. *The Annals of Statistics*, 52(2), 415–440.

Yuqian Zhang†, Abhishek Chakrabortty†, and Jelena Bradic† (2023). Double robust semi-supervised inference for the mean: selection bias under MAR labeling with decaying overlap. *In-*

formation and Inference: A Journal of the IMA, 12(3), 2066–2159.

Yuqian Zhang and Jelena Bradic (2022). High-dimensional semi-supervised learning: in search of optimal inference of the mean. *Biometrika*, 109(2), 387–403. *(This paper received the 2021 Best Student Paper Award from the ASA Nonparametric Statistics Section.)*

Zheng Zhang, Kun Zhang, Xing Yan, Songshan Yang, and **Yuqian Zhang** (2026). *Big Data in Economics and Management*. Springer Nature Singapore.

Preprints

Zhen Qi and **Yuqian Zhang*** (2026). Quadruply robust methods for causal mediation analysis. arXiv:2601.22592. Major revision at *Journal of the American Statistical Association*.

Yilizhati Anniwaer and **Yuqian Zhang*** (2025). Semi-supervised inference for treatment heterogeneity. arXiv:2509.05048. Major revision at *Biometrics*. *(Yilizhati received the Third Prize for Outstanding Papers at the 9th National Statistics Doctoral Academic Forum.)*

Kai Chen and **Yuqian Zhang*** (2025). Balancing utility and cost in dynamic treatment regimes. arXiv:2507.17360. Major revision at *Journal of the Royal Statistical Society: Series B*.

Facheng Yu, Zhen Qi, and **Yuqian Zhang*** (2024). Semi-supervised inference using unlabeled summary statistics. arXiv:2411.15691.

Yuqian Zhang and Jelena Bradic (2024). Causal inference through multi-stage learning and doubly robust deep neural networks. arXiv:2407.08560.

Yuqian Zhang, Weijie Ji, and Jelena Bradic (2024). Adaptive split balancing for optimal random forest. arXiv:2402.11228.

Awards and Honors

Junior Researcher Award, International Chinese Statistical Association	June 2025
Best Student Paper Award (Nonparametric Statistics Section), American Statistical Association	August 2021
Teaching Assistant Award, Department of Mathematics, UC San Diego	2019 – 2020
Friends of the International Center Scholarship, UC San Diego	May 2019
Outstanding Graduate, Wuhan University	June 2016

Research Grants

The NSFC General Program. "Robust causal inference under complex causal pathways". Grant No. 12671311. 2027.01–2030.12. Principal Investigator.

The RUC Endurance Research Fund for Long-Term Basic Studies. "Statistical Foundations of Artificial Intelligence". Grant No. 26XNKJ31. 2026.07–2031.06. Principal Investigator.

The NSFC Young Scientists Fund. "Statistical Inference Based on High-dimensional Semi-supervised Data". Grant No. 12301390. 2024.01–2026.12. Principal Investigator.

The RUC Start-up Fund. "Semi-supervised Inference under Selection Bias". Grant No. 23XNKJ18. 2023.01–2025.12. Principal Investigator.

The NSFC General Program. "Nonparametric Modeling and Inference for Continuous Treatment

Effect". Grant No. 12371284. 2024.01–2027.12. Co-investigator (with Zheng Zhang).

Presentations

Annual Conference of the FinTech and Big Data Technology Branch, National Society for Industrial Statistics Education and Research	Aug 2026
Frontier Forum on Bayesian Statistics and Artificial Intelligence	Aug 2026
The 4th Chinese Causal Inference Conference	July 2026
The 8th Pacific Causal Inference Conference	July 2026
The 4th Joint Conference on Statistics and Data Science & IMS-China	July 2026
Young Researchers Workshop on Statistics and Data Science	July 2026
Department of Mathematics, Hong Kong Baptist University	June 2026
IMS Asia Pacific Rim Meeting 2026	June 2026
2026 Annual Meeting of the CAAS Committee on Multivariate Analysis Application	June 2026
2026 International Conference on Frontiers of Data Science	May 2026
School of Economics and Management, Tsinghua University	Mar 2026
Sanya Workshop on Causality and Machine Learning	Jan 2026
2025 Xiamen University Symposium on Modern Statistics	Dec 2025
School of Mathematical Sciences, Shanghai Jiao Tong University	Dec 2025
2025 International Conference on Data Science and Brain-Inspired Intelligence	Dec 2025
School of Statistics & Institute of Statistics and Big Data, Renmin University of China	Sep 2025
The 8th International Conference on Econometrics and Statistics	Aug 2025
The 3rd Chinese Causal Inference Conference	July 2025
ICSA 2025 China Conference	June 2025
School of Mathematics and Statistics, Wuhan University	June 2025
2025 Annual Meeting of the CAAS Committee on Multivariate Analysis Application	June 2025
School of Management, University of Science and Technology of China	June 2025
School of Statistics, Capital University of Economics and Business	May 2025
School of Statistics & Institute of Statistics and Big Data, Renmin University of China	Apr 2025
School of Statistics, East China Normal University	Apr 2025
School of Mathematical Sciences, Tongji University	Apr 2025
School of Management, Fudan University	Apr 2025
School of Statistics and Mathematics, Central University of Finance and Economics	Mar 2025
Guanghua School of Management, Peking University	Feb 2025
School of Mathematics and Statistics, Wuhan University	Jan 2025
Causal Inference: Bridging Theory and Practice	Jan 2025
The 18th International Joint Conference CFE-CMStatistics	Dec 2024
The 1st Macau International Conference on Business Intelligence and Analytics	Dec 2024

School of Statistics and Data Science, Nankai University Dec 2024
School of Statistics and Management, Shanghai University of Finance and Economics Nov 2024
The 2nd International Conference on Machine Learning and Statistics Aug 2024
The 1st Joint Conference on Statistics and Data Science July 2024
IMS-China International Conference on Statistics and Probability July 2024
ICSA 2024 China Conference June 2024
School of Statistics and Data Science, Nankai University May 2024
School of Management, Fudan University Apr 2024
School of Statistics and Management, Shanghai University of Finance and Economics Apr 2024
The 2nd Chinese Causal Inference Conference Mar 2024
School of Statistics & Institute of Statistics and Big Data, Renmin University of China Nov 2023
IMS New Researcher Conference Aug 2023
Joint Conference on Statistics and Data Science in China July 2023
ICSA 2023 China Conference June 2023
The 1st Chinese Causal Inference Conference May 2023
School of Mathematics and Statistics, Wuhan University Apr 2023
School of Data Science, The Chinese University of Hong Kong, Shenzhen Mar 2023
Interdisciplinary Seminar on "Data Science and Life and Health" Oct 2022
Department of Applied Mathematics, The Hong Kong Polytechnic University Sep 2022
The 5th International Conference on Econometrics and Statistics June 2022
Department of Family Medicine and Public Health, UC San Diego Apr 2022
ENAR 2022 Spring Meeting Mar 2022
2021 Joint Statistical Meeting Aug 2021
Southern California Machine Learning Symposium (Poster), UC Los Angeles Mar 2019
Statistics & Data Science Symposium (Poster), UC San Diego Jan 2019

Service Activities

Council Member, Causal Inference Branch of the Chinese Association for Applied Statistics
Council Member, Chinese Statistical Association of Young Scholars
Council Member, Machine Learning Branch of the Chinese Association for Applied Statistics
Council Member, Bayesian Statistics Branch of the Chinese Association for Applied Statistics

Teaching

Instructor, Institute of Statistics and Big Data, Renmin University of China 2022 – present

- Probability Theory and Mathematical Statistics C (undergrad level) Fall 2025 – 2026
- Statistical Models and Inference II (PhD level) Spring 2026

- Machine Learning (PhD level) Fall 2025
- Causal Inference (PhD level) Fall 2024
- Special Topics in Mathematical Statistics (PhD level) Spring 2023 – 2025
- Special Topics in Business and Economic Statistics (PhD level) Fall 2023
- Special Topics in Biostatistics (PhD level) Spring 2023
- Advanced Statistical Computing (PhD level) Fall 2022

Teaching Assistant, Department of Mathematical Sciences, UC San Diego 2018 – 2022

- Math 281A, 281B – Mathematical Statistics (PhD level)
- Math 189 – Data Analysis and Inference (undergrad level)
- Math 183 – Statistical Methods (undergrad level)
- Math 180A – Introduction to Probability (undergrad level)
- Math 11 – Calculus-Based Prob & Stats (undergrad level)